

# Probability 1 – Exercises

Tutorial no. 9

13th Nov 2025

**9.1** Assume that  $Z$  is such a random variable that  $2Z$  has the same distribution. What is the distribution of  $Z$ ?

**9.2** Let  $X$  be a continuous random variable, with CDF  $F$ . What is the distribution of  $Y = F(X)$ ?

**9.3** Calculate the PDF of the following random variables:

(a) If  $\xi$  is standard normal then what is the PDF of  $X := \xi^2$ ?

**HW** (b) (1 point) If  $\xi \sim \text{EXP}(\lambda)$ , then what is the PDF of  $X := 3\xi - 2$ , and  $Z := \sqrt[4]{\xi}$ ?

(c) If  $\xi$  is uniform on the  $[-1; 1]$  interval, then what is the PDF of  $X := \xi^4$  and  $U := \sin(\pi\xi)$ ?

(d) If  $\xi$  is exponential with parameter  $\lambda$ , then what is the PDF of  $Y := e^\xi$ ?

**HW** (e) (2 points) If  $\xi$  is uniform on  $[-3; 4]$ , then what is the PDF of  $Z := \xi^2$ ?

**9.4** Let  $X$  be a standard Cauchy random variable. (Recall that its PDF is  $f(x) = \frac{1}{\pi(1+x^2)}$  for  $x \in (-\infty, \infty)$ .) Show that  $X$  and  $1/X$  have the same distribution.

**♣ 9.5** (3 point) Let  $X$  be a random variable with PDF  $\frac{1}{\ln 2} \frac{1}{1+x}$  on the  $[0, 1]$  interval, and 0 otherwise. What is the PDF of the fractional part of  $\frac{1}{X}$ ?

**9.6** Let  $\xi$  denote the distance of the point  $X$  and the point of the plane with coordinates  $(1; 1)$ . Find the probability density function of  $\xi$  if

(a)  $X$  is uniformly distributed on the  $[0, 1]$  interval of the x-axis.

(b)  $X$  is uniformly distributed on the  $[0, 2]$  interval of the x-axis.

**9.7** A stick of length  $\ell$  is broken at two independently and uniformly distributed points. What is the expected value of the length of the shortest one out of the three pieces obtained in this way?

**9.8** We roll two independent fair dice. What is the joint probability weight function of  $X$  and  $Y$  if

**HW** (a) (2 points)  $X$  is the minimum of the two throws,  $Y$  is their sum;

(b)  $X$  is the first throw,  $Y$  is the maximum of the two throws?

**9.9** From the Leaning Tower of Pisa, I drop a ball from height 40.5 meters. As we know from physics, in  $t$  seconds it falls  $gt^2/2$  meters, where  $g = 9m/s^2$ , hence it reaches the ground in exactly 3 seconds. Each of my two friends, Galileo and Isaac, takes a photo of the experiment, at independent random times with distribution  $\text{Uni}[0; 3]$  sec.

(a) In expectation, how high is the ball on Isaac's photo?

(b) In expectation, how high is the ball on the photo that is taken later?

**HW 9.10** (3 points) Jack arrives to school with a delay of  $X \sim \text{Uni}[0, 12]$  minutes. Jill arrives with a delay of  $Y \sim \text{Uni}[0, 8]$  minutes, independently of Jack.

(a) What is the probability that Jill arrives earlier than Jack?

(b) What is the expectation of the difference  $|X - Y|$ ?

**9.11** Let  $(X, Y)$  be the pair of coordinates of uniformly chosen point of the unit disc of the plane. Determine the p.d.f. of the marginal distributions.

**9.12** Let  $(X, Y)$  denote a pair of random variables with joint p.d.f.  $f(x, y)$ . Are  $X$  and  $Y$  independent?

(a)  $f(x, y) = xe^{-(x+y)} \mathbf{1}[x > 0, y > 0]$ .

(b)  $f(x, y) = 2 \cdot \mathbf{1}[0 < x < y < 1]$ .

**9.13** Let  $X$  and  $Y$  be random variables with joint probability density function

$$f(x, y) = \begin{cases} \frac{4}{5}(x + xy + y) & \text{if } 0 < x, y < 1 \\ 0 & \text{else.} \end{cases}$$

Calculate the marginals. Are the two variables independent?

**9.14** Calculate the marginal distributions! Are  $X$  and  $Y$  independent?

**HW** (a) (2 points) Let the joint probability density function of  $X$  and  $Y$  be

$$f(x, y) = \begin{cases} A \cdot (x^2y + y^2x) & \text{if } 0 < x, y < 1 \\ 0 & \text{else} \end{cases}$$

where  $A$  is a positive constant that must be calculated first.

(b) Same question when  $(X, Y)$  is uniform on the unit circle.

(c)  $(X, Y)$  is a uniform distribution on  $D = \{(x, y) \in \mathbb{R}^2 : 2x^2 + y^2/2 \leq 1\}$ .

**9.15** A rectangle has sides with length 1 and  $a$ . We choose one point on each side of length 1, with independent uniform distributions. Let  $X$  be the distance of these two points. What is the PDF of  $X$ ?